University of Batna 2
Department of English Language and Literature
Module: **Linguistics**
Second-Year BA/ Groups **6** & **7**
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Generative Linguistics

Introduction: Noam Chomsky (1928- Present)

Noam Avram Chomsky was born on December 7, 1928, in Philadelphia, Pennsylvania. He is one of the most important figures in modern linguistics and the founder of Generative Linguistics. Chomsky revolutionised the study of linguistics in the second half of the twentieth century. His work marked a turning point in the field and introduced new ways of understanding language.

Noam Chomsky (1928- Present) is a central figure in modern linguistics. His work in generative grammar transformed the discipline and marked a major shift in linguistic theory. Through his publications, teaching, and intellectual influence, he has had a lasting impact on linguistics and beyond.

1. Early Life and Education

Chomsky's parents were William Chomsky, a Hebrew scholar, and Elsie Simonofsky Chomsky. Although their first language was Yiddish, it was forbidden in their home. Instead, Chomsky grew up reading Hebrew and later taught Hebrew classes.

He first attended Oak Lane Country Day School and later Central High School in Philadelphia. In 1945, he entered the University of Pennsylvania, where he studied linguistics, mathematics, and philosophy. He earned his Ph.D. in 1955. After completing his doctorate, he spent four years as a Harvard Junior Fellow, conducting doctoral research at Harvard University.

2. Academic Career

In 1955, the Massachusetts Institute of Technology (MIT) offered Chomsky a position.

- In 1961, he became a full professor in the Department of Modern Languages and Linguistics.
- In 1976, he was appointed Institute Professor.
- He taught at MIT for over 50 years.

He taught at MIT for over 50 years. He is now Institute Professor Emeritus of Linguistics at MIT. Since 2017, he has been affiliated with the University of Arizona.

3. Contributions to Linguistics

- Chomsky has made major contributions to the study of linguistics. His work has also influenced the teaching of computer languages and mathematics.
- Some of the historical principles of linguistics that he learned early from his father helped shape his later theories. His Master's research focused on modern spoken Hebrew.
- His greatest contribution is considered to be his work on generative grammar, which developed from his interest in modern logic and mathematical foundations.

4. Major Publications

Chomsky's major linguistic works include:

- *Syntactic Structures* (1957)
- *Aspects of the Theory of Syntax* (1965)
- *The Sound Pattern of English* (1968, with Morris Halle)
- *Language and Mind* (1972)
- *The Logical Structure of Linguistic Theory* (1975)

- *Reflections on Language* (1975)

He also wrote *Language and Responsibility* (1979), which links language and politics.
His political writings include:

- *American Power and the New Mandarins* (1969)
- *The Fateful Triangle: The United States, Israel and the Palestinians* (1983; updated edition 1999)
- *Hegemony or Survival: America's Quest for Global Dominance* (2003)

5. The Linguistic Revolution

- By around 1950, linguistics had lost touch with other disciplines and had become an abstruse subject of little interest outside the field. It was ready for a revolution.
- In 1957, linguistics took a new turning point when Chomsky published *Syntactic Structures*, which revolutionised the study of linguistics.
- The book was later revised, which led to the publication of *Aspects of the Theory of Syntax*.
- Chomsky argued that Bloomfieldian linguistics was:
  - **Too ambitious:** it was unrealistic to expect to establish complete and perfect rules for extracting a full description of language from a mass of data.
  - **Too limited:** it concentrated on describing utterances that had already been spoken, rather than explaining the underlying system of language.

6. Recognition and Influence

- Chomsky has received a large number of honorary doctorate degrees from universities around the world, including the University of London and the University of Chicago.
- He delivered the Beckman Lectures at the University of California, Berkeley in 1967. In 1969, he presented the John Locke Lectures at the University of Oxford and the Sherman Memorial Lectures at the University of London.
- He is widely known not only for his work in linguistics, but also for his activism and criticism of United States policies, especially foreign policy.

Chomsky has been described as:

- "The most often cited living author."
- "One of the most respected and influential intellectuals in the world."

Chomsky's Principles: Key Ideas in Generative Linguistics

- According to Chomsky, a grammar should not simply be a list of sentences that have already been spoken. It must also explain how speakers are able to produce and understand new sentences that they have never heard before.
- Anyone who knows a language carries its grammar in their mind. This internal knowledge allows them to create and understand an unlimited number of new utterances.
- To understand Chomsky's theory, we need to examine the main principles he proposed.

1) Competence and Performance

One of Chomsky's most important distinctions is between **competence** and **performance**.

* **<u>Competence</u>** refers to the speaker's internal knowledge of language. Chomsky describes it as: "The ideal speaker-hearer's knowledge of his language." It is the complete mental system of grammatical rules that a speaker possesses.

To illustrate this idea, Chomsky gives his famous example:

Colourless green ideas sleep furiously.

Grammatically, this sentence is correct (Subject + Verb + Complement). However, it does not make logical sense. This shows that grammar and meaning are not the same thing. A sentence can be grammatically correct but semantically strange.

* **Performance** refers to the actual use of language in real situations.

In real speech, people hesitate, repeat words, make slips of the tongue, become tired, or change their minds. These are performance factors. Chomsky believes that when studying language, we should separate these accidents from the underlying system (competence).

Competence vs Saussure's Langue and Parole

Competence and performance are **not exactly** the same as Saussure's distinction between langue and parole. Saussure saw langue as a **social system** shared by a community. Chomsky, however, sees competence as **something in the mind of the individual, developing naturally as part of human growth.**

In other words, competence is ideal knowledge, while performance is real-life language use.

Grammaticality and Competence

Consider the following sentences:

1. Sarah wrote a letter.
2. A letter was written by Sarah.
3. Letter Sarah was by written a.

The third sentence is ungrammatical. Although it contains the same words as the other two sentences, it does not follow the correct grammatical rules and therefore does not form a meaningful sentence.

According to Chomsky, our internal knowledge of grammar (competence) allows us to recognise whether a sentence is grammatical or ungrammatical.

Chomsky is mainly interested in competence. This is one of the clearest differences between Structuralism and Transformational Grammar (TG). Structuralism focused only on language that had already occurred in texts. TG, however, is more concerned with the mental system that produces sentences rather than with the sentences themselves.

2) Recursiveness

Chomsky argues that language is recursive. This means that we can create an unlimited number of sentences using a limited number of rules.

For example:

My uncle's wife's daughter's friend's sister's doll's cotton dress is very beautiful.

- This sentence can continue endlessly. Even if we have never heard such a sentence before, we are still able to produce and understand it.
- Chomsky uses this example to show that language is not learned simply by habit or imitation. Instead, we generate sentences by applying grammatical rules that we have internalised.
- Although nobody explicitly teaches us to form sentences in this way, we can do so because we have mastered the rules of our language.

3) Introspection

Chomsky believes that speakers naturally know whether a sentence is grammatical or not. This knowledge comes from introspection, or linguistic intuition.

For example:

I was surprised by John's refusal to come.

→ This sentence contains internal relationships similar to: John refused to come.
→ Even though the first sentence does not clearly state this connection, speakers intuitively understand the relationship between *John* and *refusal*.

Ambiguity and Deep Meaning
Speakers can also recognise ambiguity. Consider the sentence:

Flying planes can be dangerous.

This sentence has one surface structure but at least two possible deep meanings:

→ It may mean that **the activity of flying planes is dangerous**.
→ It may mean that **planes which are flying are dangerous**.

Although the sentence looks the same on the surface, its deep structure allows for two different interpretations.

Surface Structure and Deep Structure
Chomsky distinguishes between:

- **Surface structure**: the form of the sentence as it appears and is pronounced. It determines the phonetic form.
- **Deep structure**: the underlying structure that determines meaning.

For example:
They asked him to come.
→ The sentence does not explicitly say who is expected to come. However, our intuition tells us that *he* is the subject of *to come*. This interpretation exists at the level of deep structure.

4) Innateness

Chomsky argues that human beings are biologically prepared to acquire language. We are born with an inbuilt capacity that allows us to learn language.

He disagrees with behaviourists, who believe that the mind at birth is like a blank page, shaped only by imitation and stimulus-response learning.

According to Chomsky, although the social environment is important and determines which language we acquire, the ability to acquire language itself is innate.

The language we learn will be the one spoken in our social environment, but our ability to learn it comes from within.

Nativist Theory, Creativity, and the LAD

Chomsky proposes the **Language Acquisition Device (LAD)**, a mental mechanism that allows children to detect and learn the rules of any language they hear. This explains the **creativity of language**: children do not simply imitate, but can **produce novel sentences** and **generalise grammatical rules** to new situations. The LAD ensures language learning is rapid and universal, even with limited input.

5) Transformational Generative Grammar

Transformational Generative Grammar (TG) is a theory that tries to explain:

- How human beings produce and understand language.
- How a finite brain can generate an infinite number of sentences.
- How language connects **sound** and **meaning**.

TG is not just about describing sentences ==> It tries to explain the **mental system** that makes language possible. A universal grammar (TG) model has three main characteristics:

First Characteristic: Finite Rules → Infinite Sentences

It should aim to clearly explain how a limited system such as the brain can work with a limited set of elements, such as words and structures, yet still produce an unlimited number of sentences. The model should reflect the competence of an ideal speaker, meaning it must be able to generate infinitely many sentences by applying a finite set of rules to a finite set of elements. We can briefly illustrate how this might work. For instance, suppose we have the following rules:

Simple Phrase Structure Rules

S → NP + VP

NP → Det + N

VP → V + NP

Let us assume we have the following small sets: **Nouns:** teachers, students/ **Determiners:** the, several, three/ **Verbs:** admire, follow, help.

From this limited system, we can form many different sentences:

Teachers admire students.
Students follow teachers.
Several teachers help students.
The three students admire the teachers.
Three teachers follow several students.

Even with just a small number of words and simple rules, we can already produce many sentences.

If we introduce recursion, meaning that clauses can be embedded inside other clauses, the number of possible sentences becomes unlimited:

The teachers who admire the students help the principal.
The students who follow the teachers respect the assistant who helps them.

→ This shows **productivity:** the ability to generate infinite sentences from finite means.
→ This reflects the ideal speaker's **linguistic competence**.

Second Characteristic: The Grammar Must Be Explicit

TG aims to describe the speaker's internal knowledge of grammar. Therefore, the model must be: clear, precise, and self-contained.

The rules alone must tell us whether a sentence is grammatical. For example:

✔ The **students** respect the **teacher**.

✖**Students the respect teacher the.**

Even though the words are correct, the second sentence violates the rule: S → NP + VP. → The grammar itself explains why it is unacceptable.

Third Characteristic: Three Components of Grammar

TG includes three components that mirror how humans process language.

1. Phonological Component

Deals with sounds (phonemes) and their combinations. For example: in English, the sound cluster /str/ is possible (street). But /zdl/ at the beginning of a word is impossible.

The phonological component determines what sound patterns are allowed (phonotactics).

2. Semantic Component
Deals with meaning and interpretation. For example: The table ate the sandwich. Grammatically correct. Semantically strange.
Why? Because "table" is not [+animate].
→ This shows grammar and meaning are related but not identical.

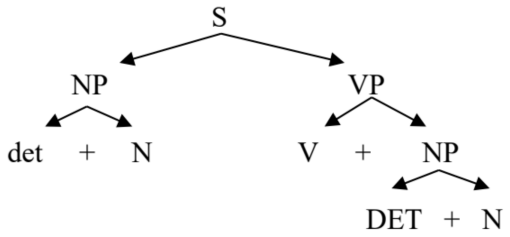
3. Syntactic Component (The Core of TG)
This is the most important part of the theory. Chomsky argues that native speakers recognise **two levels of structure**: Deep Structure and Surface Structure.

In generative grammar, a distinction is made between **deep structure** and **surface structure**. Deep structure refers to the underlying meaning and grammatical relationships in a sentence, while surface structure is the form the sentence takes after grammatical rules arrange the words. Sometimes two sentences can look very similar on the surface but have different meanings because their deep structures are different.

For example, *David is ready to help* means that David will help someone, whereas *David is easy to help* means that someone will help David. Although the sentences appear similar, their underlying meanings differ. In other cases, sentences may look different on the surface but share the same deep structure and meaning. For instance, *The teacher praised the student* and *The student was praised by the teacher* have different word orders, but they express the same basic relationship between the teacher and the student. This demonstrates that sentences can have different surface forms while conveying the same underlying meaning.

*** <u>Structure of the Syntactic Component</u>**
To explain deep and surface structure, TG divides syntax into two parts:

1. The Base Subcomponent
The base generates the **deep structure** using phrase structure rules.
Example:
S → NP + VP
NP → Det + N
VP → V + NP
Sentence:
The dog chased the cat.
Deep structure identifies:
Agent: The dog
Patient: The cat
→ This structure can be represented as a **tree diagram** (phrase marker).



2. The Transformational Component
Transformations operate on deep structure to produce surface structure.
- **Deep structure:** The dog chased the cat.
- **Transformations** produce: The cat was chased by the dog. (Passive)
The cat was chased. (Agent deletion)
The chasing of the cat (by the dog). (Nominalisation)

→ All share the same underlying pattern: Det + N + V + Det + N

*** Types of Transformational Rules**

Transformations explain how sentences change form.

Transformation	Rule	Example	Explanation
Deletion	A + B + C → A + B	Anna laughed and Mark laughed. → Anna and Mark laughed.	Repeated material is deleted.
Addition (Insertion)	A + B → A + B + C	Close the window. → You close the window.	An element is inserted.
Permutation (Movement)	A + B + C → A + C + B	Turn the light off. → Turn off the light.	Elements change position.
Substitution	A + B + C → A + D + C	Paul arrived and Lisa left. → On Paul's arrival, Lisa left.	One structure replaces another.

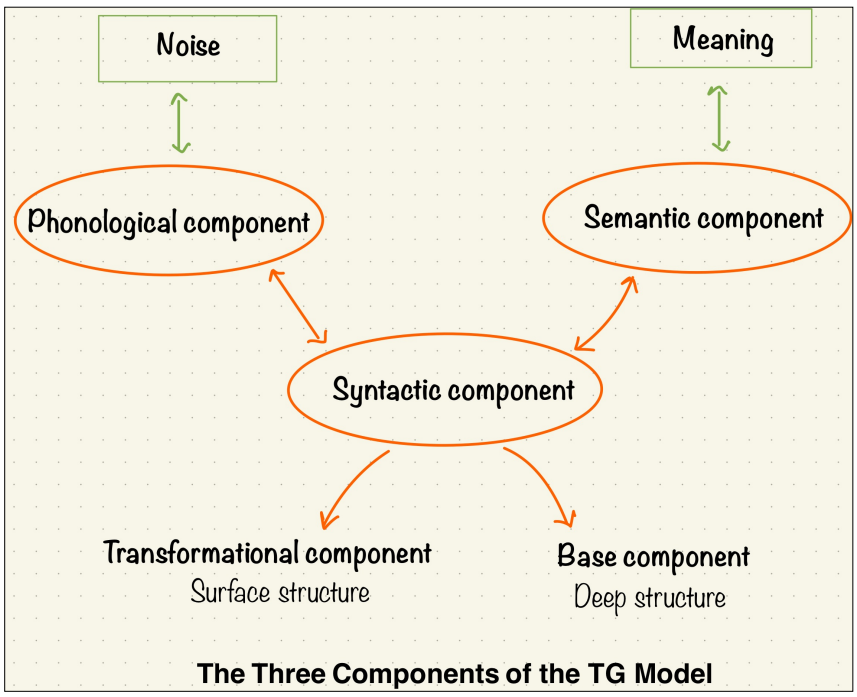
In essence, a Transformational-Generative (TG) grammar aims to link a particular sequence of sounds with a specific meaning through its syntactic rules. This relationship can be represented in a diagram, showing that the connection works both ways: a speaker can assign meaning to a string of sounds, and a listener can interpret sounds to recover meaning. Crucially, TG grammar itself is neutral, placing no preference on speaking over understanding. Its ultimate purpose is to uncover the fundamental principles of language, identifying the universals shared across all human languages, and in doing so, it offers profound insight into the workings of the human mind.

The Ultimate Aim

TG is not only about English grammar. Its broader goal is to discover **Universal Grammar:** the principles common to all human languages. By studying grammar, we study: the structure of language and the nature of the human mind.

Key Points to Remember

- A finite set of rules can generate infinite sentences.
- Grammar must be explicit and predictive.
- Language has phonological, syntactic, and semantic components.
- Sentences have deep and surface structures.
- Transformations link deep and surface forms.
- TG aims to explain universal properties of human language.



The Three Components of the TG Model